

Convolutional Neural Networks (CNN)

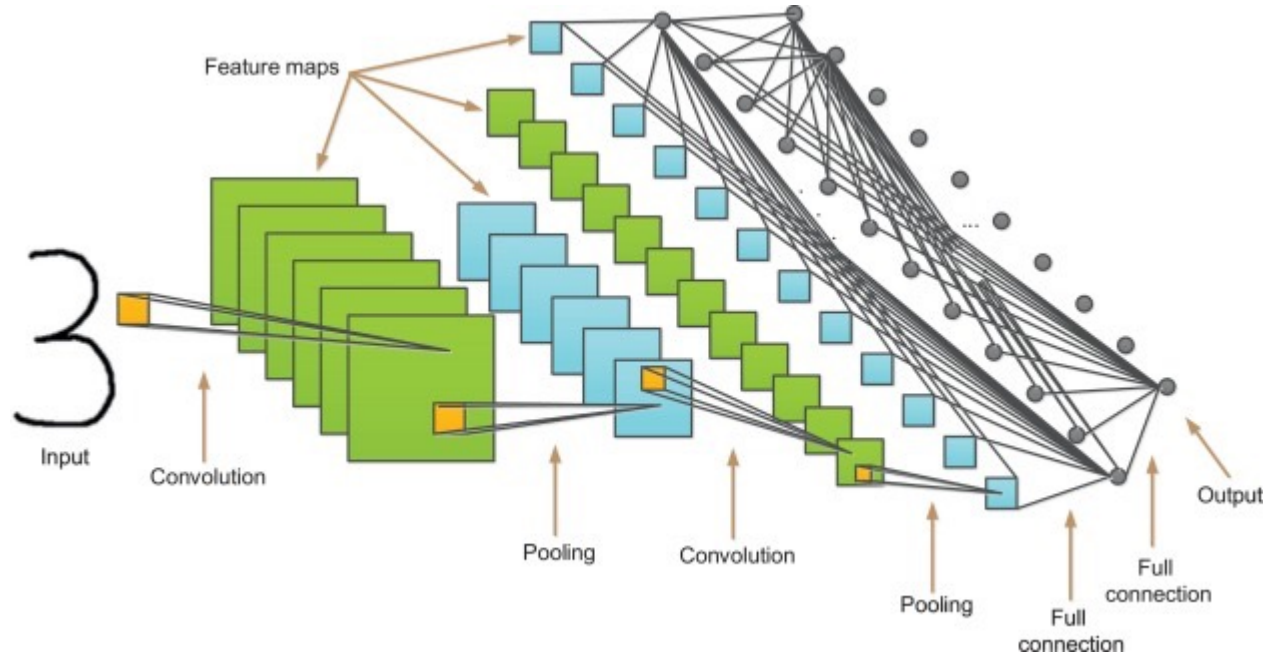
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<https://velcin.github.io>

Formation UdL 2025-2026

CNN first designed to deal with images



(source: Li et al., 2016)

Why convolutions ?

- Parameter sharing
- Sparsity of connections
- Less sensitive to transformations (e.g., translation)

Filter (aka convolution) can detect patterns

- Example of vertical lines

10	10	10	0	0	0
10	10	10	0	0	0
10	10	10	0	0	0
10	10	10	0	0	0
10	10	10	0	0	0
10	10	10	0	0	0

 $*$

1	0	-1
1	0	-1
1	0	-1

 $=$

0	30	30	0
0	30	30	0
0	30	30	0
0	30	30	0

- Different kinds of patterns



1	0	-1
1	0	-1
1	0	-1

Vertical

1	1	1
0	0	0
-1	-1	-1

Horizontal

1	0	-1
2	0	-2
1	0	-1

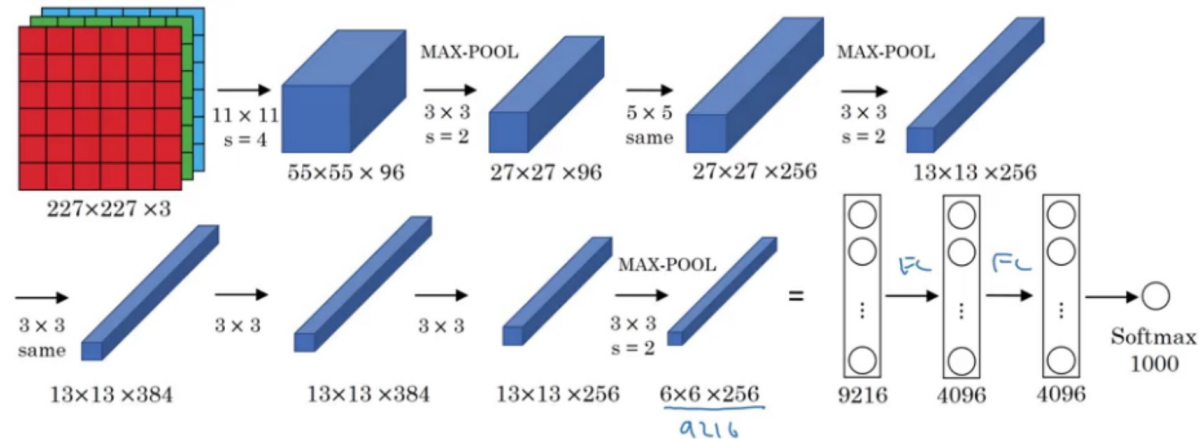
Sobel
filter

3	0	-3
10	0	-10
3	0	-3

Scharr
filter

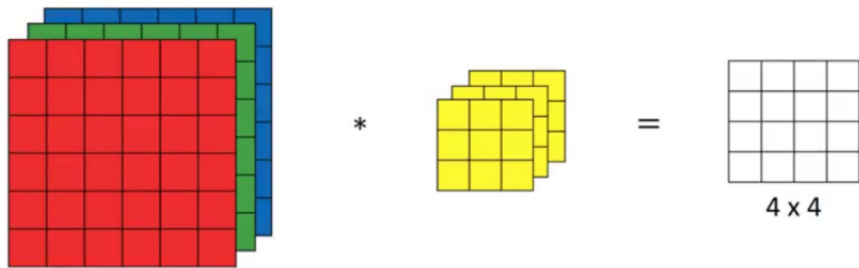
Automatically learn filters

- Example with AlexNet:



Some technical details

- One layer = *multiple* filters
- Add padding to better handle the borders
- Strided convolutions
- Convolution over volumes



CNN can be applied to other kind of data

- Textual data as 1D CNN

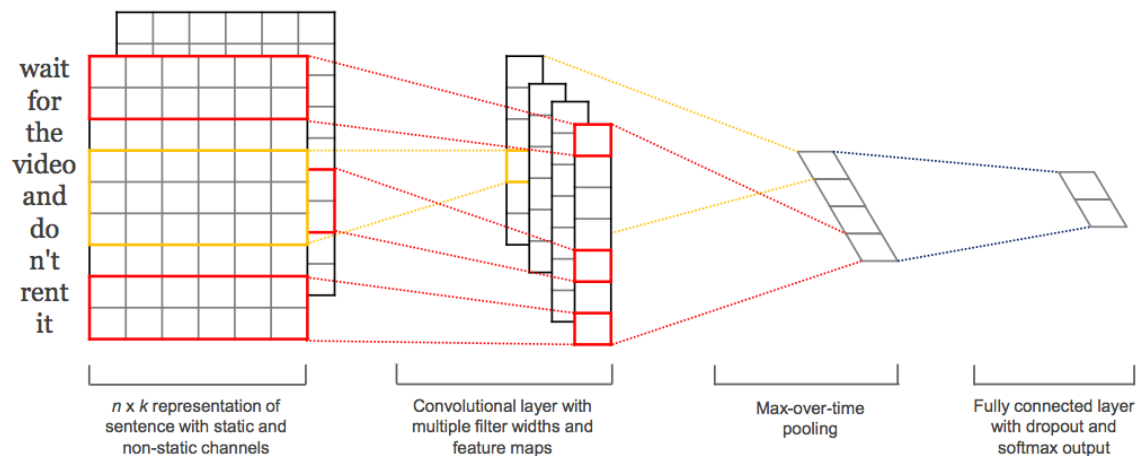


Figure 1: Model architecture with two channels for an example sentence.

(source : [https://cezannec.github.io/CNN Text Classification/](https://cezannec.github.io/CNN%20Text%20Classification/))